

MONMOUTH JET CENTER, NJ, USA

WMO: 724084

Lat: 40.183N

Lon: 74.117W

Elev: 49

StdP: 100.74

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 54760

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.8	-9.0	-22.5	0.5	-7.0	-19.8	0.7	-7.6	12.8	-0.3	11.7	-0.6	4.9	290	0.497

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.1	32.8	23.5	31.4	22.9	29.8	22.3	25.2	30.5	24.3	29.0	23.5	27.8	4.5	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.7	18.6	27.6	22.8	17.6	26.6	22.3	17.1	26.2	76.7	30.7	73.5	29.1	70.0	27.3	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.1	9.5	8.4	-15.0	35.6	3.4	1.3	-17.4	36.6	-19.3	37.3	-21.2	38.1	-23.7	39.0		
			-15.9	26.3	2.8	1.1	-17.9	27.1	-19.6	27.8	-21.1	28.4	-23.1	29.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.2	0.2	1.5	5.0	10.8	15.8	21.2	24.3	23.1	19.7	13.3	8.0	3.3
	DBStd	9.41	5.86	5.29	5.04	4.72	4.30	3.65	2.92	2.86	3.42	4.27	4.71	5.09
	HDD10.0	1089	307	242	170	42	2	0	0	0	16	93	218	
	HDD18.3	2768	562	472	414	232	105	15	1	24	165	311	467	
	CDD10.0	1908	4	4	15	67	182	336	442	408	291	118	33	9
	CDD18.3	542	0	0	1	7	27	101	184	150	64	9	0	0
	CDH23.3	4148	0	1	7	72	244	771	1569	1086	356	43	1	0
	CDH26.7	1356	0	0	1	22	74	246	588	333	86	7	0	0
Wind	WSAvg	3.8	4.4	4.5	4.7	4.5	3.6	3.4	3.1	2.9	3.3	3.6	3.9	4.2
Precipitation	PrecAvg	1208	98	69	115	95	92	102	113	117	108	101	86	109
	PrecMax	1466	201	141	252	246	204	198	178	255	212	314	170	214
	PrecMin	899	37	12	23	33	22	30	26	33	27	19	20	29
	PrecStd	132	39	32	51	47	41	45	40	52	53	68	41	45
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.4	19.1	22.6	29.3	32.0	33.6	36.0	34.0	32.1	27.4	22.1	18.9
		MCWB	13.1	13.0	14.4	18.2	21.1	23.4	24.6	25.1	23.3	19.4	16.5	15.0
	2%	DB	14.0	14.8	17.9	25.1	28.7	32.0	33.3	32.1	28.9	23.9	18.9	16.1
		MCWB	10.6	10.6	11.8	16.3	19.7	22.8	23.9	23.5	21.8	18.8	14.6	12.9
	5%	DB	11.6	12.2	15.8	21.4	26.2	29.1	32.1	30.1	27.2	22.2	17.3	13.0
		MCWB	8.7	8.5	10.3	14.3	18.3	21.6	23.3	22.9	20.4	17.7	13.3	10.3
	10%	DB	8.2	8.9	12.7	18.2	23.0	27.6	30.0	28.0	25.2	20.1	15.4	11.2
		MCWB	5.3	5.6	8.4	12.5	16.4	20.7	22.9	21.7	19.8	16.5	12.2	8.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.6	15.3	15.9	19.3	22.8	24.8	26.7	26.3	25.0	22.3	17.7	15.8
		MCDB	16.8	18.3	21.5	26.0	28.5	30.9	33.1	31.9	29.8	24.2	20.7	17.7
	2%	WB	11.8	11.5	13.4	17.5	21.1	23.8	25.3	25.1	23.5	20.3	15.7	13.7
		MCDB	13.9	13.9	16.8	23.5	26.6	29.5	31.0	29.9	26.4	23.2	17.9	15.8
	5%	WB	8.9	8.3	11.2	15.6	19.6	22.8	24.5	24.0	22.3	18.6	14.3	11.1
		MCDB	11.0	11.3	14.7	20.4	24.5	27.9	29.5	28.1	25.2	21.0	16.6	13.3
	10%	WB	6.0	5.8	8.6	13.8	17.9	21.8	23.7	23.1	21.2	16.9	12.6	8.5
		MCDB	7.9	8.7	12.7	17.8	22.1	26.3	28.2	26.6	24.2	19.4	14.9	10.8
Mean Daily Temperature Range	5% DB	MDBR	7.8	8.4	8.6	9.4	9.3	9.2	9.1	8.7	8.8	9.2	8.6	7.8
		MCDBR	10.9	12.1	12.8	13.5	13.2	11.5	11.2	10.4	10.6	11.0	10.5	9.9
		MCWBR	9.6	9.4	8.7	7.2	6.6	5.3	4.4	4.7	5.5	6.7	7.8	9.0
	5% WB	MCDBR	10.7	11.3	11.4	11.7	11.4	10.3	9.5	8.9	8.6	8.8	8.9	9.7
		MCWBR	10.0	9.7	9.1	7.4	6.2	5.5	4.4	4.4	5.2	6.5	7.9	9.6
Clear-Sky Solar Irradiance	taub	0.315	0.324	0.357	0.381	0.425	0.480	0.481	0.472	0.397	0.361	0.331	0.313	
	taud	2.482	2.460	2.412	2.373	2.282	2.166	2.175	2.193	2.374	2.480	2.519	2.534	
	Ebn at Noon	850	894	896	894	857	807	802	798	842	840	824	826	
	Edh at Noon	77	91	106	118	132	148	146	140	109	88	74	69	
All-Sky Solar Radiation	RadAvg	1.91	2.83	3.81	4.90	5.44	5.98	6.07	5.31	4.39	3.10	2.18	1.66	
	RadStd	0.17	0.29	0.32	0.45	0.59	0.54	0.29	0.35	0.42	0.30	0.19	0.16	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (61 neighbors)	+0.41	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+119	+69

Nomenclature: See separate page